Assignment 1

October 29, 2026 7:41 PM

1. Cyclic Rotation:

Define function w/ inputs array & k. the desired number of rotations

Create empty rotated list

If array is empty:

Output: rotated empty list

Else:

N = length of array

For i from 0-N:

If number of rotations is within 0-N:

Add element i-k in array to position i in rotated list

Else:

Subtract (k//N)*N from k

Add element i-k from array to position i in rotated list

Output: Rotated list

3. Frog Jump

Define function Frog Jump w/ inputs x = distance travelled, y = target distance, and d = jump distance

If (y-x) is divisible by d:

Output: (y-x)//d

Else:

Output: (y-x)//d + 1

4. Missing Element:

Define function FindMissingElement w/ input array

If array is empty:

Output: 1

Else:

Missing elements = list of integers (1-array length+1) after removing all elements in array from list

Output: Missing element minimum value

Missing elements = list of integers 1 - array length + 1 after removing all elements in array from list

Output: Missing element minimum value

7. Frog River One

Define function w/ inputs array & X = # of leaf positions

Create dictionary of fallen leaves

If removing all elements in array from a list of all integers from 1 - X creates empty list:

For element & corresponding index of array:

Add element as key w/ index as corresponding value

If # of keys = X:

Output: Maximum value in dictionary

Else:

Continue w/ next element

Else:

Output: -1

9. Missing Integer

Define function FindMissing Integer w/ input array

If Maximum value in array is less than 1:

Output: 1

Else:

Missing Integers = List of integers from 1 - array length + 1 minus all elements in array

Output: Minimum value in missing integers